



Designing a green space

Task A: Planning rooftop agriculture

- 1) The diagram shows the plan view of a rooftop for an office building. The roof has two congruent skylight windows.

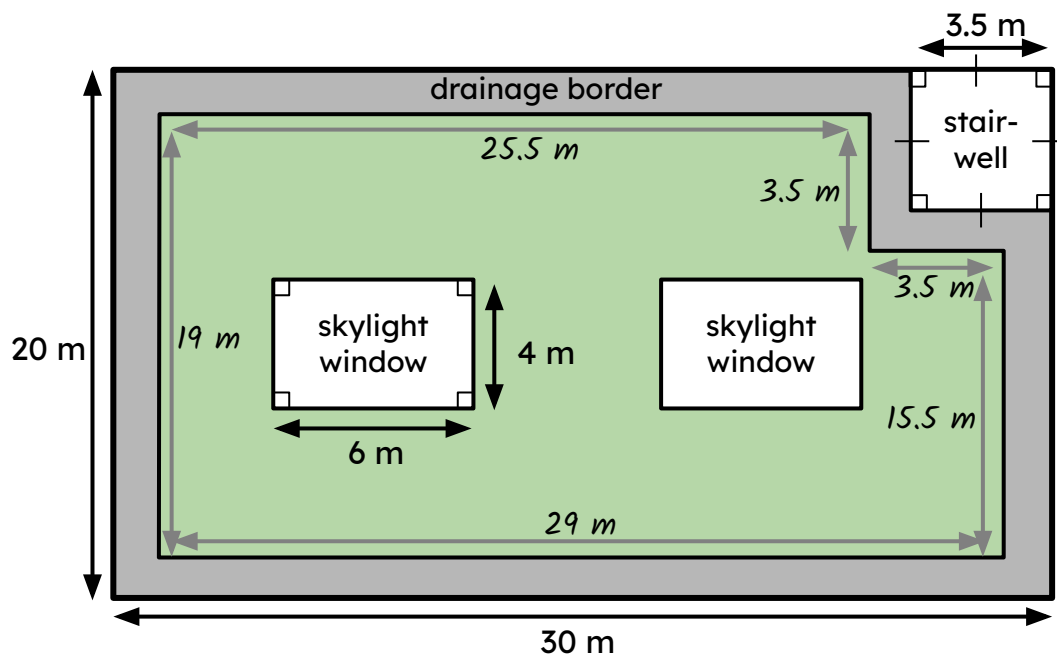


diagram not drawn to scale

A designer wants to use as much roof-space as possible to plant greenery. They need to include a border around the edge, for drainage. The border is 50 cm wide all around.

Calculate the maximum area that can be used for planting greenery.

$$\text{Area inside the border: } 25.5 \times 3.5 + 29 \times 15.5 = 538.75 \text{ m}^2$$

$$\text{Area of each window: } 4 \times 6 = 24 \text{ m}^2$$

$$\text{Area for planting greenery: } 538.75 - 2 \times 24 = 490.75 \text{ m}^2$$



Task B: Planning a garden or allotment

- 1) The diagram shows the plan for part of a community garden. It has a path that is 80 cm wide and four planting boxes. Each box is labelled with what it is intended to grow.

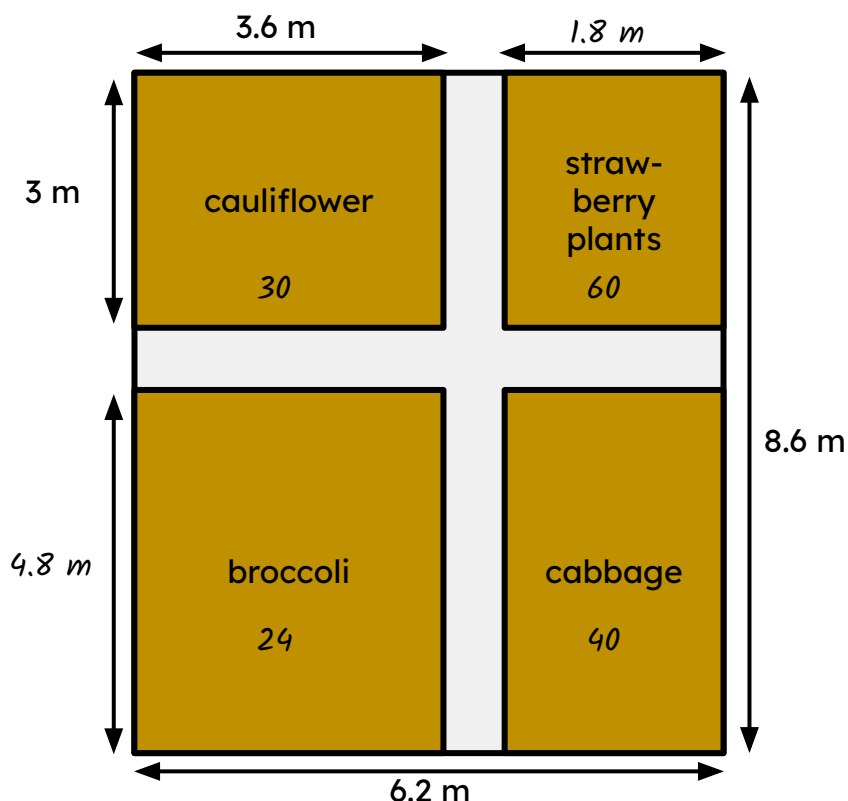


diagram not drawn to scale

The information below shows how far apart each type of plant should be:

- broccoli 80 cm
- cabbages: 45 cm
- cauliflower: 60 cm
- strawberry plants: 30 cm

What is the maximum number of plants that can be planted in each box?

Cauliflower: $360 \div 60 = 6$
 $300 \div 60 = 5$
 $6 \times 5 = 30$

Strawberry plants: $360 \div 60 = 6$
 $300 \div 60 = 5$
 $6 \times 5 = 30$

Broccoli: $360 \div 60 = 6$
 $300 \div 60 = 5$
 $6 \times 5 = 30$

Cabbage: $360 \div 60 = 6$
 $300 \div 60 = 5$
 $6 \times 5 = 30$



Task C: Designing a park

1) A city is planning to build a park in a residential area.

a) Use the information below to design its layout and construct a scale drawing.

You can either use paper (e.g. A4 or A3) or dynamic software (e.g. GeoGebra or Desmos) for your drawing. Include a scale and a key.

Information about the park:

- The space is 500 m by 600 m rectangle.
- Large trees should be planted at least 15 m apart.
- Small trees should be at least 5 m apart.
- Footpaths should be at least 2 m wide.
- A path that provides vehicle access should be at least 4 m wide.

Answers will vary. If using A4 paper, then an example scale could be 1 : 2500 (1 cm represents 25 m). Answers should include a scale and a key.

b) Plan a running route along the paths of your park. It should start and finish in the same place. Calculate the length of the route.

Answers will vary.

Example:

The perimeter of the park is 2.2 km.

Therefore, if there is a path inside the park that is close to its edges, then it could be used as a route that is slightly less than 2.2 km.

If the route also includes addition paths inside the park, then the route could be longer.

